

AI-DRIVEN MAINTENANCE REPORT

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1. Technical Sensor Data

Parameter	Measured Value
Air Temperature	300.0 °F
Process Temperature	340.0 °F
Rotational Speed	1200 RPM
Torque	65.0 Units
Tool Wear	210 %

2. AI Agent Analysis

Agent 1 (Initial Analysis):

Torque and tool wear indicate high risk due to excessive torque (65 Nm) and tool wear (210 min).

Agent 2 (Verification):

Torque >60 and tool wear >200 trigger high risk per rules.

3. Engineering Analysis & Recommendations

Summary:

REQUIRED MAINTENANCE

Analysis:

The sensor data indicates that the system is experiencing high risk due to excessive torque, tool wear, and abnormal temperatures. Specifically, the rotational speed of 1200 is within normal limits, but the air temperature of 300.0°C and process temperature of 340.0°C exceed the safety threshold by a significant margin.

Validation:

The sensor values are compared against the established thresholds as follows:

- * Torque: 65 Nm exceeds the threshold of 60 Nm
- * Tool Wear: 210 min exceeds the threshold of 200 min
- * Temperature Difference: Not applicable, but air and process temperatures exceed the safety threshold by a significant margin

Recommendation:

1. IMMEDIATELY reduce rotational speed to minimize further tool wear and torque.
2. PERFORM URGENT TOOL INSPECTION AND REPLACEMENT as necessary to prevent catastrophic failure.
3. CONDUCT A THOROUGH TEMPERATURE CONTROL CHECK to identify and correct any underlying issues causing the abnormal temperatures.

Note: These recommendations are based on the sensor data provided and may require further investigation or testing to confirm the root cause of the issue.

4. Conclusion

Final Decision: REQUIRED MAINTENANCE
Decision Source: RULE-BASED SYSTEM TRIGGERED